Lecture 11

Lobbying

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Introduction

Sources of accountability so far:

- voter's information
- politicians' time horizon

Great, but we observe political failures with

- non term limited politicians
- fairly well informed electorate (agriculture)

There must be something else

So far: citizens have only one avenue of influence, i.e., voting

In most polities, the distribution of voting rights is egalitarian

In practice, citizens

- have opposing interests
- can influence policies outside of voting
- different interests have different ability to influence

Political Economy of Trade

Why trade protection so widespread?

Theory:

Grossman and Helpman (1994)

Empirics:

Goldberg and Maggi (1999)

Rodrik and Mayda (2005)

Grossman and Helpman (1994)

Caveat: lots of moving parts

Outline

1. The economy: Consumption
2. The economy: Production
3. Political Process
4. Equilibrium Tariffs

The consumption side

$N + 1$ goods, indexed by i ($i = 0$ is money)

L households, indexed by h, each with 1 unit of labor

Each household values consumption (c) in each good:

$$U^h = c_0 + \sum_{\{1,\ldots,N\}} u_i(c_i)$$

Each household maximizes U^h subject to budget

$$I^h = \sum_{\{0,\ldots,N\}} p_i c_i$$

[normalization: $p_0 = 1$; also $\mathbf{c} = [c_1, \ldots, c_N]$]

The consumption side

Normalization: $p_0 = 1$, hence

$$c_0 = I^h - \sum_1^N p_i c_i$$

H chooses $\mathbf{c} = [c_1, ..., c_N]$ as follows:

$$\max_{\mathbf{c}} \left\{ I^h - \sum_1^N p_i c_i + \sum_1^N u_i(c_i) \right\}$$

FONC

$$u'(c_i) = p_i$$
$$\Rightarrow d_i(p_i) = u'^{-1}(p_i)$$

Demand functions

Constrained maximization yields demand function

$$d(\mathbf{p}) = [d_1(p_1), ..., d_N(p_N)]$$

N.B: d_i same across households (and only depends on p_i):

$$c_0^h = I^h - \sum_{i \in \{1,...,N\}} p_i d_i(p_i)$$

Indirect utility (induced preferences) is then

$$V(\mathbf{p}, I^h) = I^h + s(\mathbf{p}),$$

where $s(\mathbf{p})$ is surplus:

$$s(\mathbf{p}) = \sum_{i \in \{1,...,N\}} \Big(u_i[d_i(p_i)] - p_i d_i(p_i) \Big)$$

Demand functions - Example

Suppose that for each i, $u_i(c_i) = \beta c_i - \frac{c_i^2}{2}$

Then demand function is $d_i(p_i) = \beta - p_i$

Indirect utility is then

$$V(\mathbf{p}, I^h) = I^h + \sum_{i \in \{1,\ldots,N\}} \frac{(\beta - p_i)^2}{2}$$

The production side

Normalize productivity (and thus wage) in numeraire to 1

In each sector, K_i units of the specific factor of production

In each sector, inputs are labor and a specific factor:

$$y_i = f_i(L_i, K_i)$$

Assume f_i has constant returns to scale ($f_i(aL, aK) = af_i(L, K)$)

Define profit functions as

$$\begin{aligned}\pi_i(p_i) &= \max_{L_i}\{f_i(L_i, K_i)p_i - L_i\} \\ &= y_i(p_i)p_i - L_i\end{aligned}$$

Supply functions are then $y_i(p_i) = \pi_i'(p_i)$

Nota bene

Firms

- Choose quantities (Cournot)
- Not prices (Bertrand)

Wage is fixed to one, so they make positive profits

Not an explicit model of competition in each sector

Prices and tariffs

International price (exogenous) is p_i^*, such that $p_i = p_i^* + t_i$

If good i is imported (exported), t_i is a tariff (export subsidy)

(Net) imports are

$$m_i(p_i) = d_i(p_i)L - y_i(p_i)$$

Trade balance obtained through import (or export) of numeraire.

Tariff revenue (given back to consumers) is

$$T(\mathbf{p}) = \sum_{\{1,\ldots,N\}} (p_i - p_i^*)m_i(p_i)$$

Ownership structure

Specific factor in each sector owned by H_i people

Define $H = \sum H_i$; $L - H$ people do not own capital (proletariat)

Joint utility of H_i owners of K_i is

$$W_i(\mathbf{p}) = \underbrace{\pi_i(p_i)}_{\text{Profits}} + \underbrace{H_i}_{\text{Labor income}} + \frac{H_i}{L} \underbrace{T(\mathbf{p})}_{\text{Tariff revenue}} + \underbrace{s(\mathbf{p})}_{\text{Surplus}} H_i$$

Proletariat's is

$$W_0(\mathbf{p}) = \underbrace{L - H}_{\text{Labor income}} + \frac{L - H}{L} \underbrace{T(\mathbf{p})}_{\text{Tariff revenue}} + \underbrace{s(\mathbf{p})}_{\text{Surplus}} (L - H)$$

Aggregating yields $W(\mathbf{p}) = \sum W_i(\mathbf{p})$

Political process

$i \in O$ (organized) or $i \in U$ (unorganized)

There is a government that cares about contributions ($\sum C_i$) and welfare

Each organized sector $i \in O$ maximizes payoff $W_i - C_i$, chooses a contribution schedule

$$C_i(\mathbf{p})$$

Govt chooses $\mathbf{p}$ in order to maximize

$$G(\mathbf{p}) = \sum_{i \in O} C_i(\mathbf{p}) + \alpha W(\mathbf{p})$$

α measures how much govt internalizes citizens' welfare

Equilibrium Tariffs I

Equilibrium analysis from Bernheim and Whinston (1986)

Focus on *truthful contribution schedules*:[1] organized sector i contributes $C_i(\mathbf{p}) = \mathbf{max}\{0, W_i(\mathbf{p}) - B_i\}$

Can rewrite govt's objective as

$$G(\mathbf{p}) = \sum_{i \in O}[(1+\alpha)W_i(\mathbf{p}) - B_i] + \sum_{i \notin O} \alpha W_i(\mathbf{p})$$

Essentially, govt maximizes a *weighted welfare function*

1. The constants $\mathbf{B} = [B_1, \ldots, B_N]$ are not unique, focus on equilibria in which constraint does not bind.

Equilibrium Tariffs II

$\lambda = \sum_{i \in O} H_i / L$ population share owning capital in an organized industry

$\varepsilon_i = -\frac{\partial m_i}{\partial p_i} \frac{p_i}{m_i} > 0$ is the *elasticity* of (net) imports to price:

$\eta_i = m_i / y_i$ is the import penetration in i

FONC imply

$$\frac{t_i}{p_i} = \begin{cases} \frac{1-\lambda}{(\alpha+\lambda)\eta_i \varepsilon_i} & \text{if } i \in O \\ -\frac{\lambda}{(\alpha+\lambda)\eta_i \varepsilon_i} & \text{otherwise} \end{cases}$$

Wait, what??

$$\frac{\partial G(\mathbf{p})}{\partial p_i} = \sum_{j \in O}[(1+\alpha)\frac{\partial}{\partial p_i}W_j(\mathbf{p})] + \sum_{j \notin O}[\alpha\frac{\partial}{\partial p_i}W_j(\mathbf{p})]$$

$$\frac{\partial}{\partial p_i}W_j(\mathbf{p}) = \begin{cases} \pi_i'(p_i) + \frac{H_i}{L}\frac{\partial}{\partial p_i}T(\mathbf{p}) + H_i\frac{\partial}{\partial p_i}s(\mathbf{p}) & if i = j \\ \frac{H_j}{L}\frac{\partial}{\partial p_i}T(\mathbf{p}) + H_j\frac{\partial}{\partial p_i}s(\mathbf{p}) & i \neq j \neq 0 \\ \frac{L-H}{L}\frac{\partial}{\partial p_i}T(\mathbf{p}) + (L-H)\frac{\partial}{\partial p_i}s(\mathbf{p}) & j = 0 \end{cases}$$

$$\frac{\partial}{\partial p_i}T(\mathbf{p}) = m_i(p_i) + t_i\frac{\partial m_i(p_i)}{\partial p_i} = d_i(p_i)L - y_i - t_i\varepsilon_i\frac{m_i(p_i)}{p_i}$$

$$\frac{\partial}{\partial p_i}s(\mathbf{p}) = -d_i(p_i) + \frac{\partial d_i(p_i)}{\partial p_i}\underbrace{[u_i'[d_i(p_i)] - p_i]}_{=0} = -d_i(p_i)$$

Wait some more

Use $\pi_i'(p_i) = y_i$

$$\frac{\partial G(\mathbf{p})}{\partial p_i} = \sum_{j \in O}[(1+\alpha)\frac{\partial}{\partial p_i}W_j(\mathbf{p})] + \sum_{j \notin O}[\alpha\frac{\partial}{\partial p_i}W_j(\mathbf{p})] = 0$$

$$0 = y_i \begin{cases} (1+\alpha) & i \in O \\ \alpha & \text{otherwise} \end{cases}$$

$$+ (1+\alpha)\left[\lambda\frac{\partial}{\partial p_i}T(\mathbf{p}) + \lambda L\frac{\partial}{\partial p_i}s(\mathbf{p})\right]$$

$$+ \alpha\left[(1-\lambda)\frac{\partial}{\partial p_i}T(\mathbf{p}) + (1-\lambda)L\frac{\partial}{\partial p_i}s(\mathbf{p})\right]$$

And $\frac{\partial}{\partial p_i}T(\mathbf{p}) + L\frac{\partial}{\partial p_i}s(\mathbf{p}) = -y_i - t_i\varepsilon_i\frac{m_i(p_i)}{p_i}$

And some more

$$\frac{\partial G(\mathbf{p})}{\partial p_i} = \sum_{j \in O} [(1+\alpha) \frac{\partial}{\partial p_i} W_j(\mathbf{p})] + \sum_{j \notin O} [\alpha \frac{\partial}{\partial p_i} W_j(\mathbf{p})] = 0$$

$$0 = y_i \begin{cases} (1+\alpha) & i \in O \\ \alpha & \text{otherwise} \end{cases}$$

$$+ [(1+\alpha)\lambda + \alpha(1-\lambda)] \Big[\frac{\partial}{\partial p_i} T(\mathbf{p}) + L \frac{\partial}{\partial p_i} s(\mathbf{p}) \Big]$$

$$t_i \varepsilon_i \frac{m_i(p_i)}{p_i} = -y_i + \frac{y_i}{\lambda + \alpha} \begin{cases} (1+\alpha) & i \in O \\ \alpha & \text{otherwise} \end{cases}$$

$$\frac{t_i}{p_i} \varepsilon \eta_i = \frac{1}{\lambda + \alpha} \begin{cases} 1+\lambda & i \in O \\ -\lambda & \text{otherwise} \end{cases}$$

Conceptually, this is what GH do

1. Simplified, but "reasonable" model of
   - Supply side of economy (yields $\pi(\mathbf{p})$)
   - Demand side of economy (yields $s(\mathbf{p})$)
2. A clear view of the policy making process
   - Some sectors can influence policy (contributions)
   - Govt respons to contributions (but also to welfare)
   - By setting t_i, govt effectively sets domestic price
3. Assumptions make govt's choice of $\mathbf{p}$ tractable
   - Clear empirical predictions

So what?

$$\frac{t_i}{p_i} = \begin{cases} \frac{1-\lambda}{(\alpha+\lambda)\eta_i\varepsilon_i} & \text{if } i \in O \\ -\frac{\lambda}{(\alpha+\lambda)\eta_i\varepsilon_i} & \text{otherwise} \end{cases}$$

1. Sectors gets tariff if organized, import subsidy if not
2. If $\lambda = 1$ no protection (free trade)
3. If λ declines, organized sectors get *more* protection
4. Higher protection in organized sectors with lower import penetration (!)
5. But effect of import penetration depends on whether i is organized

Intuition

Unorganized sectors get import subsidy to compensate for distortions in organized sectors

Larger import penetration increases the welfare loss from trade protection: T is less responsive to p_i (i.e., revenues grow *less* due to dead weight loss) when η_i is larger

Higher elasticity $\rightarrow$ more distortions for given tariff

What the data say - 1

Sectors gets tariff if organized, import subsidy if not

Obviously inconsistent with data

BUT, if assume that ownership is super-concentrated, $\lambda \sim 0$

$$\frac{t_i}{p_i} = \begin{cases} \frac{1}{(1+\alpha)\eta_i\varepsilon_i} & \text{if } i \in O \\ 0 & \text{otherwise} \end{cases}$$

Testing the theory

Maggi and Goldberg, AER 1999

Take the model, find empirical analog for each variable

$$\frac{t_i}{p_i} = \frac{1-\lambda}{(\lambda+\alpha)\eta_i\varepsilon_i}$$

Compute implied coefficients

$$\frac{t_i}{1+t_i}e_i = \frac{I_i - \alpha_L}{\frac{\beta}{1-\beta}+\alpha_L}\frac{X_i}{M_i} + \epsilon_i$$

$$= \gamma\frac{X_i}{M_i} + \delta I_i\frac{X_i}{M_i} + \epsilon_i$$

[Third term: main contribution wrt previous work]

Maggi and Goldberg, AER 1999

$$Tariff_i = const. + \gamma \frac{X}{M_i} + \delta \frac{X}{M_i} I_i + u_i$$

(N.B. X_i/M_i in their notation is $1/\eta_i$ in our notation)

Problems:

- Measurement in dependent variable
- Measurement of political contributions
- What about ε's?

Brillant solutions

$$(4) \quad y_i^* = \frac{t_i^* e_i}{1 + t_i^*} = \gamma \frac{X_i}{M_i} + \delta I_i \frac{X_i}{M_i} + \epsilon_i$$

$$(5) \quad t_i = \begin{cases} \frac{1}{\mu} t_i^* & \text{if } 0 < t_i^* < \mu \\ 0 & \text{if } t_i^* \leq 0 \\ 1 & \text{if } t_i^* \geq \mu \end{cases}$$

$$(6) \quad \frac{X_i}{M_i} = \zeta_1' \mathbf{Z}_{1i} + u_{1i}$$

$$(7) \quad I_i^* = \zeta_2' \mathbf{Z}_{2i} + u_{2i}$$

$$(8) \quad I_i = \begin{cases} 1 & \text{if } I_i^* > 0 \\ 0 & \text{if } I_i^* \leq 0 \end{cases}.$$

Methodological interlude - 1

Why move elasticity on LHS?

Because measurement error

- in independent variable leads to biased coefficients
- less serious concern in dependent variable

Methodological interlude - 1

$$Y = \beta_0 + \beta_1 X^* + \varepsilon$$

but can only observe $X = X^* + w$,

where $\mathrm{E}[w] = 0$, $\mathrm{V}ar[w] = \sigma_w^2$

OLS on

$$Y = \beta_0 + \beta_1 X + [\varepsilon - \beta_1 w]$$

Is biased, since

$$Cov[X, \varepsilon - \beta_1 w] = \mathrm{E}[(\varepsilon - \beta_1 w)(w)] = -\beta_1 \sigma_w^2$$

Methodological interlude - 2

How do they estimate the model?

Maximum Likelihood....what?

Basic idea is compute probability distribution of $L(X(n), \theta)$

- where $X(n)$ is a sample
- θ is a parameter
- and find set of parameters that maximize likelihood of that dataset

What the data say - 2

If $\lambda = 1$ *no protection*

This is hard to test

GM test for a "smoother" version:

TABLE A3—RESULTS FROM AN EXTENDED SPECIFICATION
($\mu = 1$)

Variable	Coefficient	Standard error	t
X_i/M_i	−0.0092	0.0044	−2.104
$(X_i/M_i) * I_i$	0.0089	0.0089	1.998
Constant	−0.2545	0.2409	−1.057
I_i	0.3851	0.3466	1.111

Note: Dependent variable: $(t_i^* e_i / 1 + t_i^*)$.

What the data say - 3

If λ declines, organized sectors get more protection

No direct evidence, use Mayda and Rodrik (05):

conjecture that inequality increase protectionism

based on estimated effect of income (rich are pro trade)

micro implications not fully inconsistent with GH...

...but macro implication that inequality $\nearrow$ protectionism is

What the data say - 4

Protection is higher in sectors with lower import penetration

GM: weak support (How can we see that from table?)

Not controlling for political organization, you get opposite!

TABLE 1—RESULTS FROM THE BASIC SPECIFICATION (G-H MODEL)

Variable	$\mu = 1$	$\mu = 2$	$\mu = 3$
X_i/M_i	−0.0093	−0.0133	−0.0155
	(0.0040)	(0.0059)	(0.0070)
$(X_i/M_i) * I_i$	0.0106	0.0155	0.0186
	(0.0053)	(0.0077)	(0.0093)
Implied β	0.986	0.984	0.981
	(0.005)	(0.007)	(0.009)
Implied α_L	0.883	0.858	0.840
	(0.223)	(0.217)	(0.214)

What the data say - 5

Effect of import penetration depends on whether i is organized

GM find Strong support:

TABLE 1—RESULTS FROM THE BASIC SPECIFICATION (G-H MODEL)

Variable	$\mu = 1$	$\mu = 2$	$\mu = 3$
X_i/M_i	−0.0093	−0.0133	−0.0155
	(0.0040)	(0.0059)	(0.0070)
$(X_i/M_i) * I_i$	0.0106	0.0155	0.0186
	(0.0053)	(0.0077)	(0.0093)
Implied β	0.986	0.984	0.981
	(0.005)	(0.007)	(0.009)
Implied α_L	0.883	0.858	0.840
	(0.223)	(0.217)	(0.214)

Bonus

Estimated α is huge

In other words, less protection than we would fear

Ways out: upstream and downstream industrites

Opposing interests within same industry

Offsetting effects